

Jesus D. Medina Novelo

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Software Engineer

Summary

Technology Engineer with 4+ years of experience developing software, AI-powered solutions, business automation, and cloud-based applications. Specialized in designing and building Full-Stack applications, ERP systems, CRM platforms, and SaaS products using Python, React, TypeScript, and modern cloud architectures. Experienced in integrating LLMs, OCR, AI agents, APIs, and machine learning workflows to optimize business operations, automate processes, and accelerate digital transformation. Strong background in AWS, Docker, PostgreSQL, Linux infrastructure, and AI-assisted software development.

Core Skills

Programming Languages - Python · JavaScript · TypeScript · SQL

Frontend & Full Stack Development - React 19 · TypeScript · Vue.js · Nuxt · TanStack Start · TanStack Router · TanStack Query · Tailwind CSS v4 · shadcn/ui · HTML · CSS

Backend y APIs - Flask · Django · FastAPI · REST APIs · API Integrations · Microservices Architecture

Artificial Intelligence & Automation - OpenAI API · LLMs · AI Agents · OCR · Chatbots · Prompt Engineering · Workflow Automation · n8n · Botpress

Automation & Integration (Low-Code/No-Code) - n8n · Botpress · Ecosistema Zoho (CRM, Inventory, Books) · OpenAI API · OCR · Chatbots

Cloud y DevOps - AWS · Vercel · DigitalOcean · Docker · Kubernetes · Linux · CI/CD

Databases - PostgreSQL · MongoDB · Supabase · SQL

Enterprise Platforms - Zoho CRM · Zoho Inventory · Zoho Books · ERP Development · CRM Development · SaaS Development

Tools - Git · GitHub · Supabase Auth · Supabase Storage · React Hook Form · Zod · FlutterFlow

Experience

Head of IT & Software Engineering

Feb 2026 – Present

Diprolam | MTY, GTO, PUE & TIJ

- Lead the company's technology operations, ensuring the availability, reliability, and continuity of critical systems, infrastructure, and business services across multiple locations.
- Manage the development of Full-Stack applications that automate inventory management, purchase orders, and administrative workflows, significantly reducing manual effort and improving operational visibility.
- Architect and develop a corporate ERP platform to centralize inventory control, procurement, warehouse management, supplier operations, and administrative processes, enhancing operational efficiency and decision-making.
- Design and build a multi-tenant SaaS platform focused on business process digitalization and automation, including authentication, file storage, document generation, and role-based access control.
- Develop and implement an enterprise CRM solution to streamline customer management, sales pipeline tracking, opportunity management, and internal business workflows.
- Deliver software and automation solutions that eliminate repetitive tasks, increase team productivity, and reduce errors associated with manual processes.
- Manage cloud infrastructure, networking, and deployment environments using Linux, Docker, AWS, and DigitalOcean, ensuring scalability, stability, and operational security.
- Design and implement cybersecurity policies, backup strategies, and disaster recovery plans (DRP) to protect critical business assets and minimize operational risks.
- Collaborate directly with administrative, operational, and engineering teams to identify improvement opportunities and transform business requirements into high-impact technology solutions.

Technologies: Python, React 19, TypeScript, TanStack Start, TanStack Router, TanStack Query, Tailwind CSS v4, shadcn/ui, Supabase (PostgreSQL, Auth, Storage), Zod, React Hook Form, Docker, AWS, DigitalOcean, Linux, REST APIs, GitHub, Vercel.

AI & API Software Developer

Jan 2025 – Feb 2026

Diario de Yucatán | Mérida

- Lead the company's IT operations, infrastructure management, and software engineering initiatives, ensuring high system availability and operational reliability.
- Develop full-stack applications and internal platforms to automate inventory, purchasing, and administrative workflows, significantly reducing operational workload.
- Design and implement backend systems, API integrations, and automation solutions to improve engineering processes, operational efficiency, and data accuracy.
- Manage cloud infrastructure, deployment environments, networking, and cybersecurity practices using Linux, Docker, AWS, and DigitalOcean.
- Implement security policies, backup strategies, and disaster recovery plans (DRP) to protect critical business systems and infrastructure.

Technologies: Technologies: Python, Flask, PostgreSQL, React, Tailwind CSS, Linux, Docker, AWS, DigitalOcean, REST APIs, SQL, Automation

AI Solutions Engineer & Project Lead

Ago 2025 – Ene 2026

Grupo Megamedia | Mérida

- Led the coordination and implementation of AI-driven solutions and automation projects across multiple business areas.
- Designed and integrated conversational AI systems, backend services, and workflow automations to optimize operational processes and improve efficiency.
- Collaborated with cross-functional teams to identify AI adoption opportunities, define technical requirements, and support digital transformation initiatives.
- Implemented API integrations, intelligent workflows, and scalable backend solutions using modern AI and cloud technologies.
- Evaluated emerging AI tools and automation platforms to improve productivity, operational scalability, and business decision-making.

Technologies: Technologies: Python, FastAPI, REST APIs, OpenAI API, LLMs, Workflow Automation, Docker, PostgreSQL, React, Cloud Infrastructure, AI Automation, n8n

Web Developer

Jan 2024 – Dec 2024

Diario de Yucatán | Mérida

- Developed an intelligent chatbot for Empléate.app, automating registration and application processes, improving user experience.
- Implemented AI-based automation to detect promoted job vacancies and assist applications, increasing conversion rates.
- Led development and improvement of the Hemeroteca portal, integrating AI features for optimized search and historical content access.
- Implemented OCR-based intelligent search, improving document information retrieval.
- Optimized performance and stability of web platforms, reducing errors and improving load times.

Technologies: Nuxt, Vue.js, Vuetify, MongoDB, JavaScript, OCR, AI, Chatbase, Anthropic Claude, REST APIs, HTML, CSS, WordPress, WooCommerce, Zoho

Software Engineering Intern

Sep 2023 – Jan 2024

Universidad Anáhuac Mayab | Mérida

- Contributed to the development of institutional web applications using React, TypeScript, and Nuxt.
- Participated in the creation of several academic websites, improving features and user experience.
- Implemented performance improvements and continuous maintenance on web platforms.

Technologies: React, TypeScript, JavaScript, HTML, CSS

Technical Support / Bot Developer

Feb 2023 – Sep 2023

Universidad Anáhuac Mayab | Mérida

- Provided technical support resolving user issues through ticketing systems and direct assistance.
- Developed a Discord bot using Python and OpenAI API for automation and community management.
- Improved response times and efficiency in technical problem resolution.

Technologies: Python, OpenAI API, Discord Bot, REST APIs, Git, Ticketing Systems

Education

Universidad Anáhuac Mayab

Bachelor's Degree in Information Technology and Digital Business

- Web Application Development
 - Software Testing & Quality Assurance Methodologies
 - Advanced Design & Database Optimization
 - Data Analysis & Intelligent Data Processing
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Certifications & Courses

- Software Development with Claude Code
- Advanced AI-Assisted Coding for Developers
- Designing Machine Learning Workflows in Python
- AI Engineer for Developers Associate
- AWS Cloud Practitioner (CLF-C02)
- CCNA – Cisco
- AWS Cloud Development
- SOC – Cisco
- Google Business Intelligence
- Data Structures in Python
- Generative AI with Large Language Models – DeepLearning.AI
- Machine Learning with Python – Coursera
- Docker & Kubernetes: The Practical Guide
- Full-Stack Web Development with React
- Advanced Prompt Engineering
- Associate Cloud Engineer – Google Cloud
- Introduction to Artificial Intelligence – Data Analyst Path (GCP)
- Data Analyst Learning Path
- Data Structures and Algorithms
- Database Management Essentials
- Packet Tracer
- SAP Professional Fundamentals